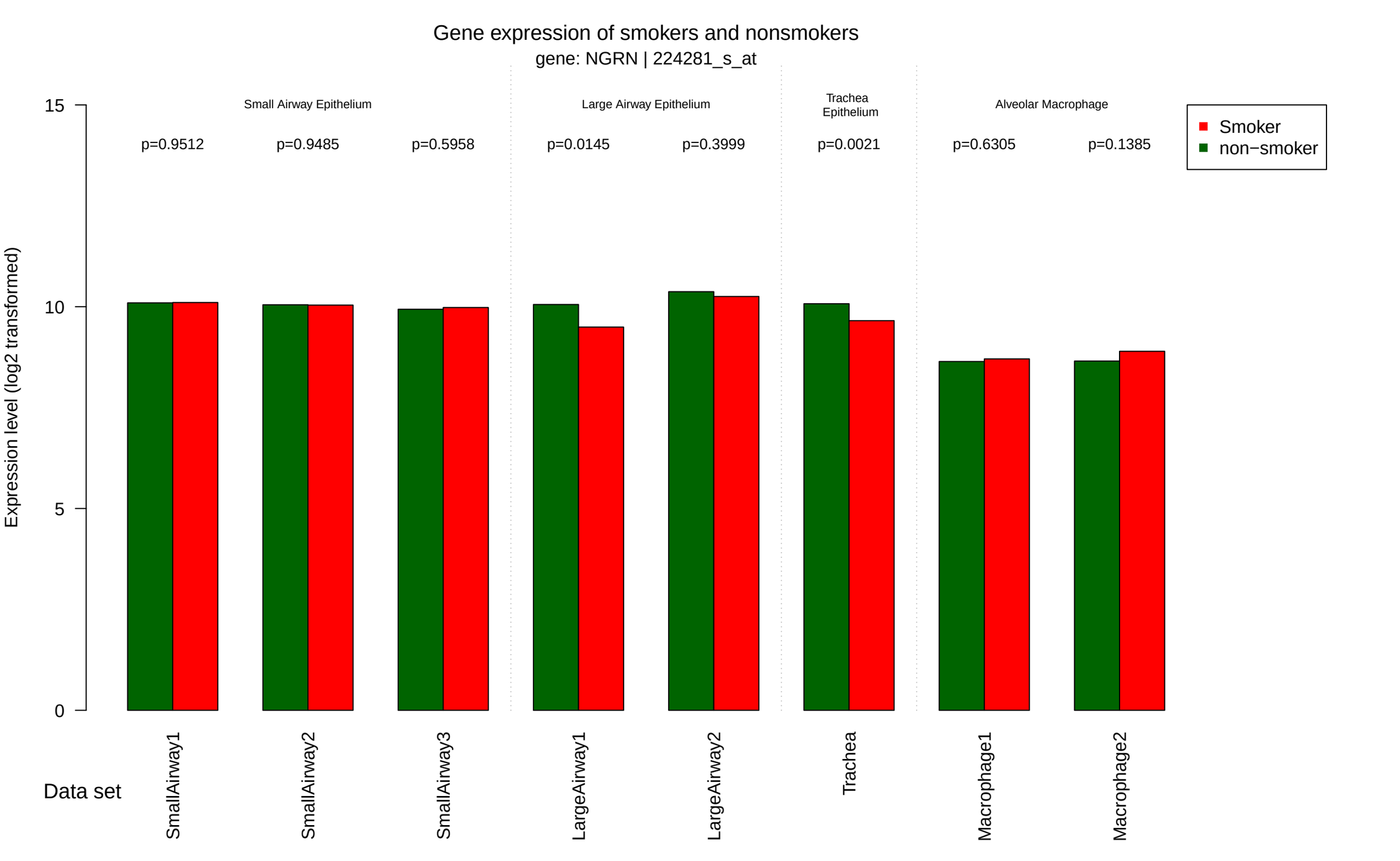
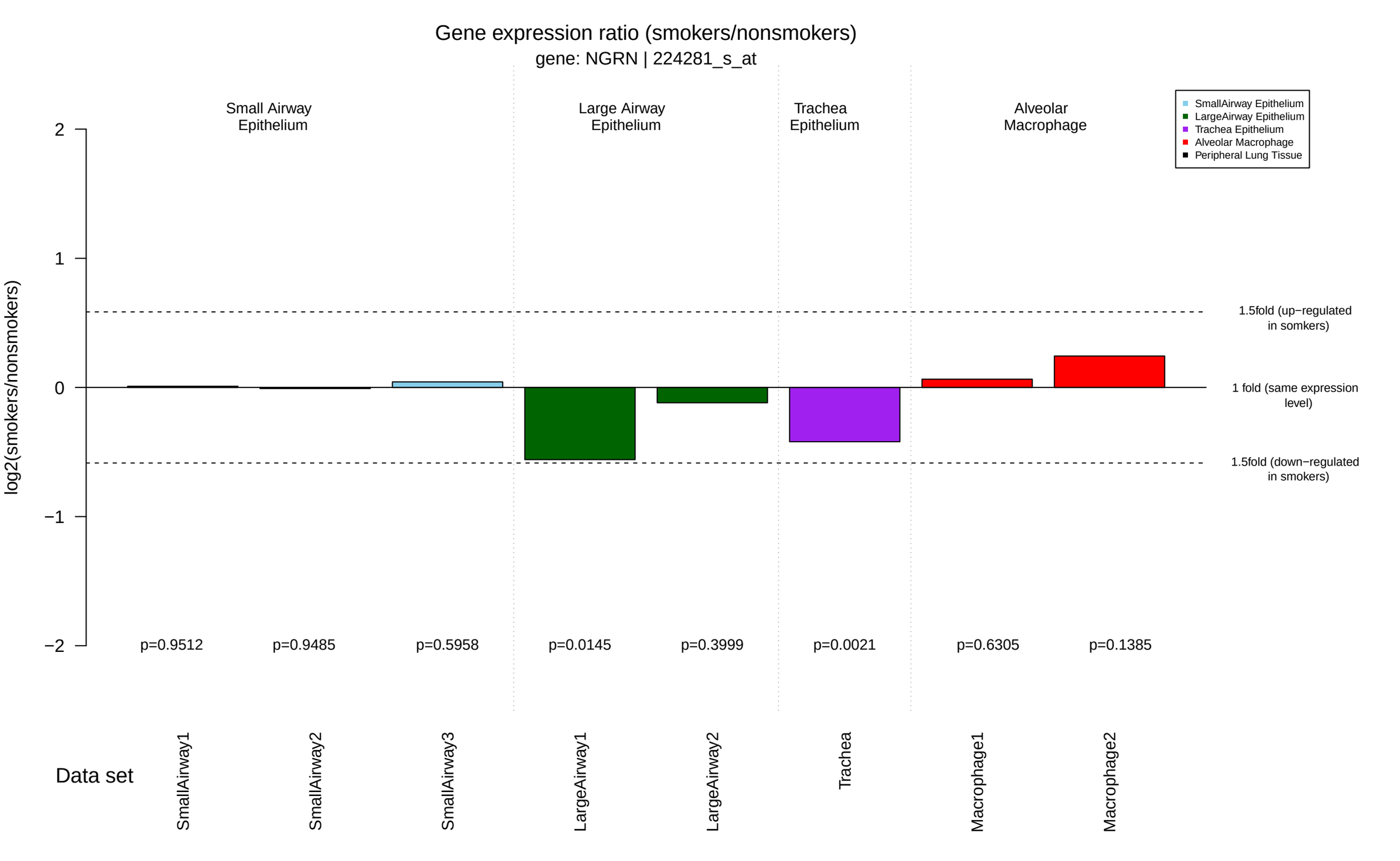
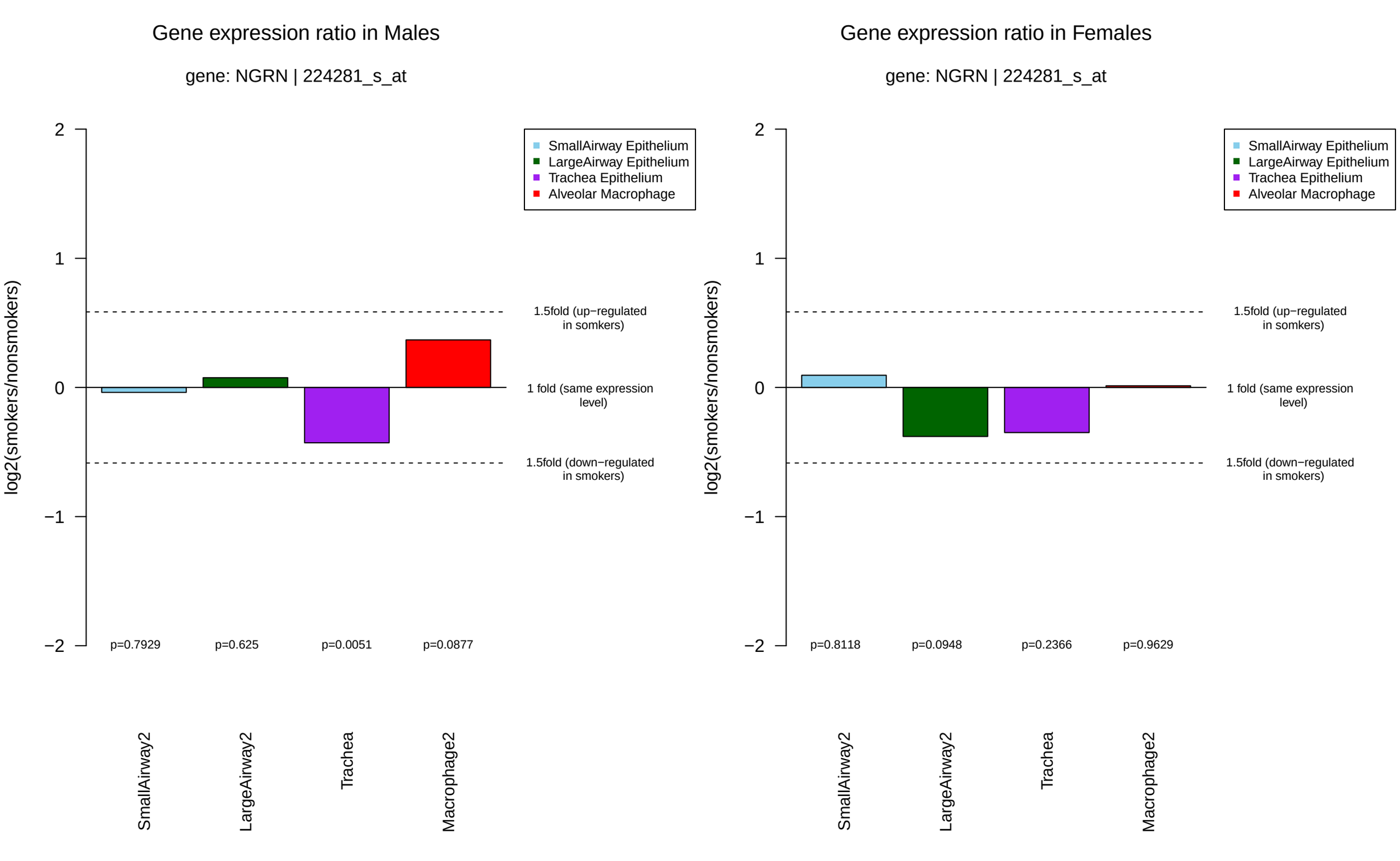
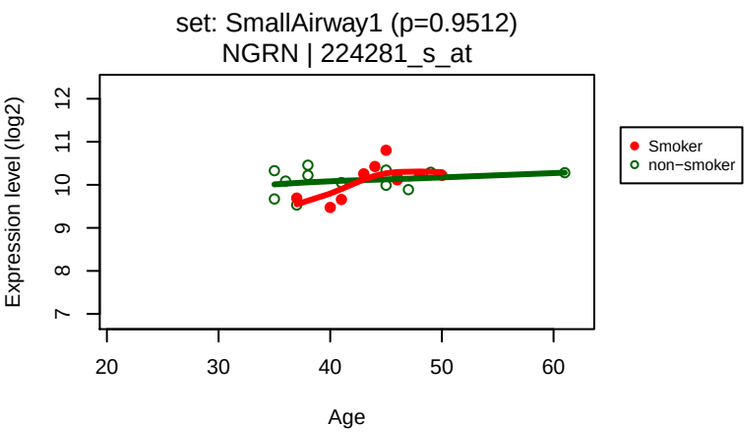
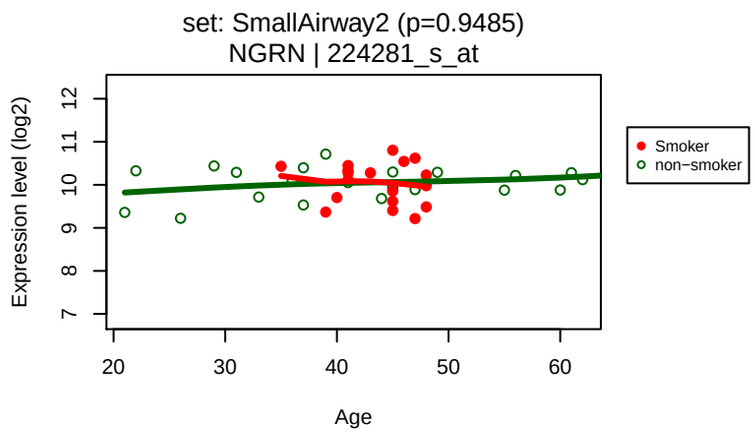
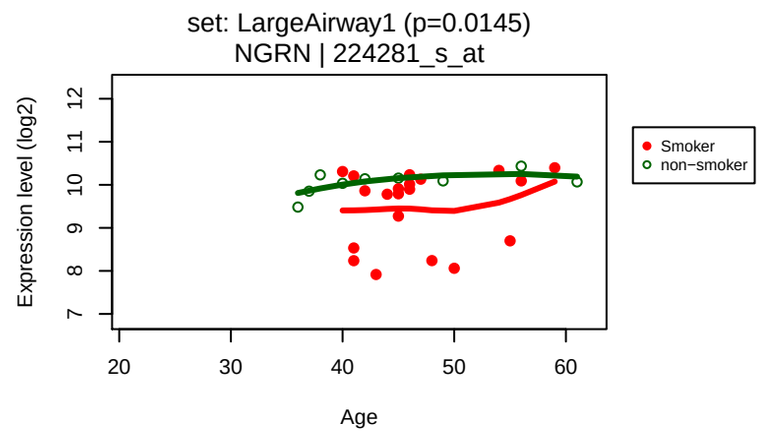

Gene expression of smokers and nonsmokers
gene: NGRN | 224281_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.9512
p=0.9485
p=0.5958
p=0.0145
p=0.3999
p=0.0021
p=0.6305
p=0.1385
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

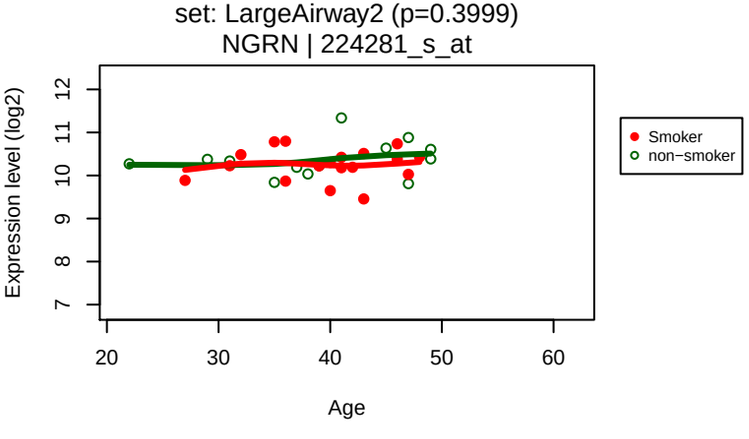

Gene expression ratio (smokers/nonsmokers)
gene: NGRN | 224281_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9512
p=0.9485
p=0.5958
p=0.0145
p=0.3999
p=0.0021
p=0.6305
p=0.1385
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

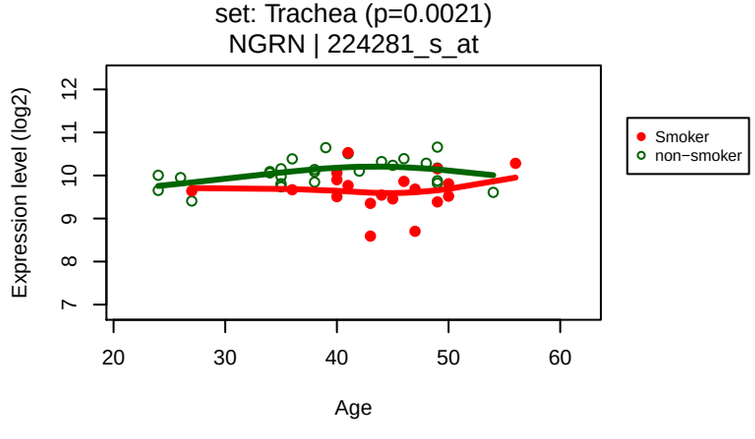

Gene expression ratio in Males
gene: NGRN | 224281_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7929
p=0.625
p=0.0051
p=0.0877
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: NGRN | 224281_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8118
p=0.0948
p=0.2366
p=0.9629
SmallAirway2
LargeAirway2
Trachea
Macrophage2

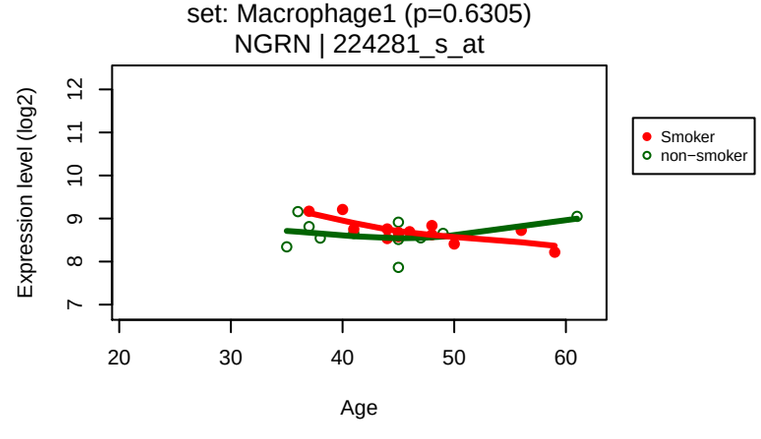

set: SmallAirway1 (p=0.9512)
NGRN | 224281_s_at
Expression level (log2)
Age
Smoker
non−smoker

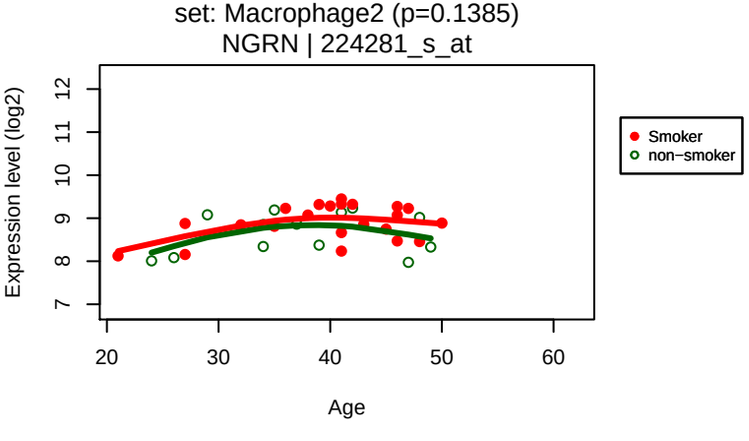

set: SmallAirway2 (p=0.9485)
NGRN | 224281_s_at
Expression level (log2)
Age
Smoker
non−smoker

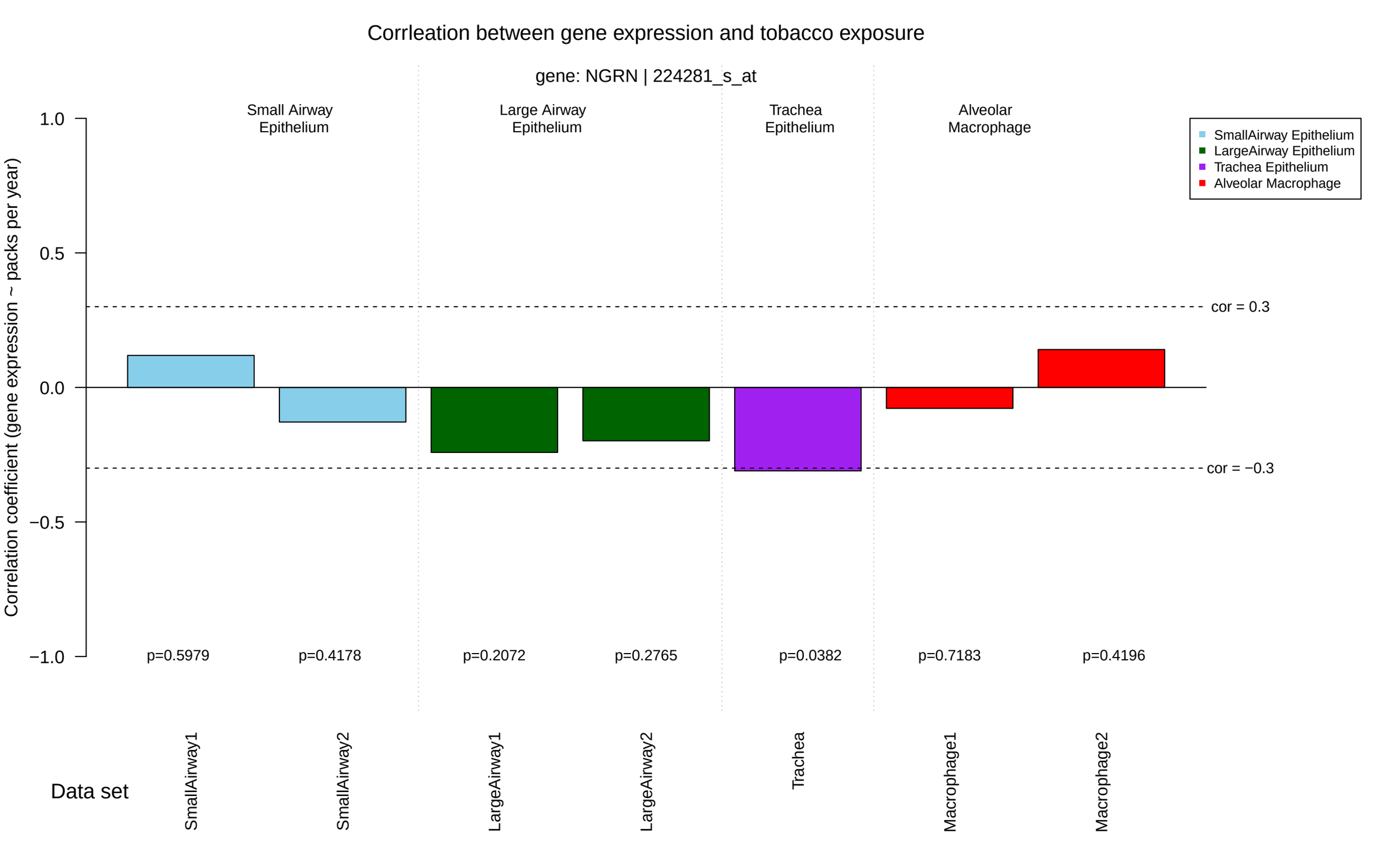

set: LargeAirway1 (p=0.0145)
NGRN | 224281_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.3999)
NGRN | 224281_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0021)
NGRN | 224281_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.6305)
NGRN | 224281_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.1385)
NGRN | 224281_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: NGRN | 224281_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.5979
p=0.4178
p=0.2072
p=0.2765
p=0.0382
p=0.7183
p=0.4196
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2